RIPPLE

Developer-Ready Software Requirements Specification (SRS)

Explainable Credit Risk Assessment for SMEs Using Alternative Data and Graph Neural Networks

Purpose: This document converts the project idea into a buildable specification for the developer. It defines what the system should do, what the two user types should see, what data should be handled, and where the developer has technical freedom.

Project phase	Phase II + Phase III combined for the 30 September milestone
Primary user	Bank / lender credit-risk personnel: credit officer, analyst, or equivalent
Secondary user	SME / MSME applicant or business user
Primary role of RIPPLE	Decision-support credit-risk assessment; final lending decision remains with the responsible officer
Core technologies	Python, FastAPI, React.js, PostgreSQL, XGBoost, PyTorch/PyG, SHAP, NetworkX, Git/GitHub

Important: Technical choices marked “Developer discretion” are intentionally not fixed. The developer may choose the most appropriate implementation, provided the required business behaviour and outputs are preserved.

1. Executive Specification

RIPPLE is a credit-risk decision-support application for SME/MSME lending. Its central purpose is to reduce information asymmetry: SMEs may have incomplete conventional credit histories, while lenders may have insufficient reliable information to assess repayment risk. RIPPLE should therefore combine conventional financial analysis with alternative, verifiable business information and relationship/network information.

The system has two user experiences. The **bank-side experience is the complete A-to-Z application**: search/select an SME, ingest data, validate data quality, analyse financial and behavioural indicators, analyse business relationships, generate model-based risk outputs, explain the result, run scenarios, and produce a credit-risk report. The **SME-side experience is intentionally smaller**: the business can submit selected information, obtain a credit-readiness/risk score or report, identify missing/weak areas, and understand what can be improved.

RIPPLE should provide its own recommendation if the developer implements that feature, but the system must clearly position the recommendation as **decision support rather than an autonomous lending decision**. The responsible bank officer retains final discretion.

The build should maximise analytical depth without indiscriminately collecting every possible variable. The developer should prefer reliable, relevant and verifiable data, reduce redundant features, and document data provenance.

The project is an academic prototype. Real production banking integration, legally binding lending decisions and uncontrolled use of private financial data are outside the present scope.

2. Users and Access Model

User	Primary purpose	Required capabilities
Bank / Credit Officer / Analyst	Full credit-risk assessment	Create/select SME, ingest files, validate data, view financial analysis, view network, run
SME / MSME Applicant	Credit-readiness and self-assessment	Enter/upload selected data, generate score/report, see missing-data alerts, understand
Developer / Admin (internal prototype role)	System and model maintenance	Manage datasets, retrain models, inspect model metrics, manage configuration; exact U

Authentication, role-based access control and admin capabilities may be implemented at the level appropriate for the academic prototype. Exact authentication technology is **Developer discretion**.

3. Core User Journey

The following is the intended end-to-end journey. This is the most important functional flow in the SRS.

```
BANK USER
   ↓
Login / open dashboard
   ↓
Search or select SME
   ↓
Create new assessment OR open previous assessment
   ↓
Upload / enter available data
   ■■■ Standard financial/tax file → automatic extraction
   ■■■ Structured CSV/data file     → automatic mapping
   ■■■ Manual input                 → only for missing/required fields
   ↓
Data validation + provenance + completeness check
   ↓
Financial & alternative-data feature engineering
   ↓
Business relationship / graph construction
   ↓
 ■■■
```

```
                    ↓
PD + risk category + risk indicators
                    ↓
 SHAP / model explanation layer
                    ↓
 Network view + key risk exposures
                    ↓
 Scenario / sensitivity analysis
                    ↓
   RIPPLE recommendation
                    ↓
Officer reviews and makes final decision
                    ↓
    Save assessment + report
```

SME JOURNEY

```
SME opens self-assessment
   ↓
Enter/upload selected business information
   ↓
Automatic extraction where possible
   ↓
Missing / weak-data alerts
   ↓
Credit-readiness / risk assessment
   ↓
Score + major positive/negative drivers
   ↓
Improvement guidance
   ↓
Optional downloadable/shareable report
```

4. Functional Requirements — Bank Side

FR-01 — SME Search and Profile

- The bank user shall be able to search for an SME and open its profile.
- The profile should show identity/basic business information, industry, business age, location and available financial information.
- The system should show whether data is verified, estimated, missing, uploaded by the user, or obtained from another permitted source.
- The user should be able to create a new assessment or open an existing assessment.

FR-02 — Data Ingestion

- The system shall accept structured files such as CSV/Excel and suitable standard-format financial/tax files where feasible.
- Where a source follows a predictable structure, the system should automatically map fields to the internal schema.
- The user should not be forced to manually enter information that can be reliably extracted from an uploaded structured source.
- If required information is absent, the system shall identify the missing field and request manual input where appropriate.
- The exact parser/extraction method is Developer discretion.

FR-03 — Data Validation and Reliability

- The system shall validate data types, ranges, consistency and basic accounting/financial logic where applicable.
- Each important data element should have a source/provenance indicator where feasible.
- Reliable/verified data should receive higher trust than unverified user-entered information.
- Clearly invalid or unverifiable data should be rejected, quarantined, or excluded according to the validation policy.

- The system should not silently treat missing or low-quality data as genuine financial facts.

FR-04 — Financial and Alternative-Data Analysis

- The system shall calculate relevant financial, behavioural and business-risk indicators.
- The system should use both direct financial information and alternative signals where reliable data is available.
- The feature set must be relevance-driven; collecting more variables is not automatically better.
- Redundant or highly overlapping features should be controlled to reduce noise and data-mining bias.

FR-05 — Credit-Risk Prediction

- The system shall produce an estimated probability of default or equivalent risk output.
- XGBoost should be evaluated as the primary tabular candidate, alongside suitable baseline models.
- The developer may change the final model architecture if validation demonstrates a better approach, but the comparison and justification must be documented.
- The model shall not be treated as infallible; uncertainty, data quality and limitations should be visible.

FR-06 — Business Network Analysis

- The system shall represent relevant business relationships as a graph where appropriate.
- SMEs/business entities should be represented as nodes and relevant business relationships as edges.
- The GNN shall be used to capture relational/network information that borrower-level tabular features may not capture.
- Exact node features, edge semantics, edge features, directed/undirected representation, GNN architecture and graph-learning objective are Developer discretion.
- The UI should provide a network view understandable to a non-ML user.

FR-07 — Risk Fusion / Model Comparison

- The system shall compare borrower-level financial risk with network-informed risk.
- The final fusion method is Developer discretion and must be empirically justified.
- At minimum, the system/report should make it possible to distinguish what came from the tabular branch and what came from the network branch.
- The project should test whether network information adds predictive value beyond financial/alternative data alone.

FR-08 — Explainability

- The system shall explain why an individual risk assessment was generated.
- SHAP should primarily be used to explain the XGBoost/tabular prediction.
- The explanation should identify major risk-increasing and risk-reducing factors in plain language.
- GNN-specific explainability is optional/Developer discretion; standard SHAP should not be assumed to explain the GNN automatically.

FR-09 — Scenario and Sensitivity Analysis

- The system should avoid relying only on a single point estimate.
- The user should be able to view risk under multiple prediction horizons where the available data supports it, such as 9, 12, 18 and 24 months.
- The system should support sensitivity/scenario analysis around material financial or business assumptions where technically feasible.
- Scenarios may include changes in revenue growth, cash-flow stability, payment behaviour, leverage, customer concentration or other material drivers.
- The exact scenario engine is Developer discretion, but outputs must clearly state that they are scenario estimates rather than guaranteed future outcomes.

FR-10 — Recommendation

- The system may provide a RIPPLE recommendation such as lower-risk / review / higher-risk or approve/review/reject.
- The developer has discretion over whether to implement a direct approve/reject recommendation.
- If implemented, the UI must clearly state that the bank officer retains final decision authority and may accept or override the recommendation.
- The recommendation should be traceable to the underlying risk assessment and not presented as a legally binding decision.

FR-11 — Assessment History and Reporting

- Completed assessments should be saved with timestamp, input snapshot/reference, model/version, outputs and explanation.
- The bank user should be able to view previous assessments.
- The system should generate a concise credit-risk report containing key financial indicators, PD/risk, network findings, major drivers, data-quality warnings and scenario outputs.

5. Functional Requirements — SME / MSME Side

The SME experience is intentionally not a full mirror of the bank application. It should provide useful self-service functionality without exposing internal lender controls or unnecessary model details.

Feature	What the SME sees
Business profile	Basic business information and selected financial inputs
Data upload	Ability to upload supported structured/standard files where feasible
Missing-data alert	Fields/information that could not be obtained or verified
Credit-readiness score	A simple score/risk band, with clear disclaimer that it is an indicative assessment
Key strengths	Factors that currently support stronger creditworthiness
Key weaknesses	Factors contributing negatively to the assessment
Improvement guidance	Practical areas the business can improve, e.g. payment consistency or cash-flow stability
Report	Simple downloadable/shareable summary if implemented

The SME interface should **not** expose confidential lender-side information, internal bank comments, hidden portfolio information, or sensitive details about other businesses.

6. Data Specification

The following is the intended data universe. The developer should implement only fields that can be obtained reliably and that contribute meaningfully to credit-risk analysis.

6.1 Business / Identity Data

- Business name and identifier where legitimately available
- Business type/legal structure
- Industry/sector
- Business age
- Geographic information
- Promoter/management information where appropriate and lawful
- Business model / operating characteristics

6.2 Financial Statement Data

- Revenue / sales and revenue growth
- Cost structure and operating expenses
- EBITDA/operating profit where available
- Net profit
- Cash and cash equivalents
- Accounts receivable and payable
- Inventory
- Current and non-current assets/liabilities
- Total debt / borrowings
- Equity/net worth
- Interest expense
- Operating cash flow / free-cash-flow-related measures where available

6.3 Credit-Analysis Ratios and Derived Indicators

- Liquidity: current/quick-type measures where data permits
- Leverage: debt-to-equity, debt-to-assets or comparable measures
- Coverage: interest-coverage / debt-service-related measures where inputs exist
- Profitability: margins, ROA/ROE-type measures where appropriate
- Efficiency/activity: receivable, payable and inventory turnover / days
- Cash-flow adequacy and stability
- Trend and growth measures

These categories align selectively with the credit-analysis portions of the CFA curriculum, particularly financial-statement analysis, ratio analysis, cash-flow analysis, credit risk, probability of default, and profitability/leverage/coverage assessment. CFA Institute describes corporate credit analysis as using business-model assessment, financial statements, cash-flow projections and profitability, leverage and coverage metrics to assess creditworthiness. ■cite■turn0search10■turn0search3■

6.4 Alternative Data

- Transaction frequency and volume patterns
- Average transaction size and variability
- Cash-in/cash-out behaviour where legitimately available
- Payment timeliness and payment consistency
- Revenue/transaction trend and volatility
- Supplier concentration
- Customer/buyer concentration
- Business relationship stability
- Tax/GST-related information where legally accessible and suitable
- Public corporate/regulatory filings and company information where legitimately accessible
- Other verifiable digital/operational signals if a defensible relationship to credit risk can be established

The project should encourage **novel alternative-data thinking**—for example, external operational indicators that indirectly reveal business activity—but a signal must not be included merely because it is unusual. It should have a defensible economic rationale, reliable provenance, and a validation path. Satellite imagery, footfall, parking utilisation or similar signals are examples of possible future research directions, not mandatory current inputs.

6.5 Relationship / Network Data

- Supplier relationships
- Customer/buyer relationships
- Transaction relationships
- Relationship duration or strength where available
- Transaction volume/frequency where available
- Other business-to-business links relevant to network exposure

The exact graph schema is Developer discretion, but the source and meaning of every relationship should be documented.

7. Data Quality, Missing Data and Verification Policy

This is a critical part of RIPPLE. The system should not compensate for poor information by pretending that the information is accurate.

Condition	Expected behaviour
Verified/reliable data	Use normally and retain provenance
User-entered but unverified data	Use only where appropriate; label as unverified
Missing but non-critical field	Continue assessment and show missing-data warning
Missing material field	Request user input where feasible and quantify/flag its impact
Large amount of material data missing	Do not generate a misleading risk result; assessment may be rejected/inconclusive
Invalid/inconsistent data	Reject, quarantine or request correction
Conflicting sources	Prefer the higher-confidence source according to documented hierarchy; flag the conflict

A **Data Completeness / Reliability indicator** should be visible in the bank-side result. A high-risk result caused by genuinely weak business fundamentals must not be confused with a high-risk result caused primarily by missing information.

8. Financial Analysis Layer

The financial-analysis layer should convert raw data into decision-useful indicators before ML modelling. CFA Institute describes financial analysis as collecting, processing, analysing/interpreting data, communicating conclusions and following up; it also notes that financial analysis supports credit-risk assessment. ■cite■turn0search2■turn0search3■

The system should therefore examine at least five dimensions:

1. **Liquidity:** ability to meet near-term obligations.
2. **Solvency / leverage:** debt burden relative to capital, assets or earnings.
3. **Profitability:** ability to generate sustainable operating and net profits.
4. **Coverage / repayment capacity:** ability to service interest and debt obligations.
5. **Cash-flow quality and stability:** whether accounting performance is supported by cash generation and whether cash flows are volatile.

Additional dimensions should include revenue growth, working-capital behaviour, receivable/payable trends, concentration risk, business-model risk, management/operational considerations where data permits, and industry/business-cycle context where a reliable dataset exists.

The application should show these indicators in a human-readable form rather than exposing raw model features only.

9. Risk Model Specification

9.1 Tabular Model

- Primary candidate: XGBoost.
- Baseline candidates: Logistic Regression and Random Forest, or other suitable baselines chosen by the developer.
- Handle class imbalance appropriately.
- Perform train/validation/test separation without leakage.
- Use appropriate preprocessing and feature selection.
- Evaluate discrimination and calibration, not accuracy alone.

9.2 Graph Model

- Construct a business graph from valid relationship data.
- Use PyTorch Geometric or another suitable graph-learning framework.
- The GNN should learn network-informed representations/risk.
- The developer may choose GCN, GraphSAGE, GAT or another justified architecture.

- The graph design should be documented: nodes, edges, directionality, node features, edge features, labels and learning objective.

9.3 Combined Assessment

- Compare tabular-only performance with graph-only and/or combined approaches where data permits.
- The final fusion strategy is developer discretion.
- The project should answer whether relational information materially improves risk assessment.
- Avoid claiming that the GNN is inherently better; select the approach based on evidence.

9.4 Evaluation

- ROC-AUC
- PR-AUC, especially if defaults are imbalanced
- Precision, recall and F1
- Confusion matrix
- Probability calibration / reliability where feasible
- Performance by relevant subgroups or segments where sample size permits
- Ablation/comparison: financial features only vs. financial + alternative data vs. financial + network information

10. PD Horizons, Scenarios and Sensitivity

RIPPLE should avoid presenting one probability as the complete truth. Where the dataset supports the necessary labels/time structure, the system may show multiple horizons such as 9, 12, 18 and 24 months.

View	Purpose
Base assessment	Current estimated risk under the available information
Horizon comparison	Compare estimated risk across different time horizons
Sensitivity analysis	Show how risk changes when a material input changes
Scenario analysis	Evaluate defined business/financial situations such as revenue decline or higher payment delays

Scenario outputs must be labelled as **model-based estimates**. They must not be presented as guaranteed forecasts or causal proof that changing one variable will produce a specific real-world outcome.

11. Explainability and User-Facing Result

A typical bank result screen should look conceptually like this:

```
------------------------------------------------------------
SME: ABC Manufacturing
Assessment date: 30 Sep 2026

RIPPLE RISK ASSESSMENT
PD: 18.7%          Risk: MEDIUM-HIGH
Data confidence: GOOD

FINANCIAL HEALTH
Liquidity       ■■■■■■■■■■
Leverage        ■■■■■■■■■■
Profitability   ■■■■■■■■■■
Cash-flow       ■■■■■■■■■■
Payment trend   ■■■■■■■■■■

TOP RISK DRIVERS
↑ High cash-flow volatility
↑ High customer concentration
↑ Recent payment delays

TOP SUPPORTING FACTORS
```

```
↓ Stable revenue growth
↓ Strong interest coverage

NETWORK VIEW
12 connected entities | 2 high-risk exposures
[Interactive business graph]

SCENARIOS
9M     12M     18M     24M
...    ...     ...     ...

RIPPLE RECOMMENDATION
"Further credit review recommended"

Officer decision:
[Approve] [Reject] [Further Review]
------------------------------------------------------------
```

The exact visual design is developer discretion, but the information hierarchy should remain understandable to a bank analyst within seconds.

12. Application UI / Wireframe Specification

The following wireframes are logical wireframes, not pixel-perfect design. They define what must exist on each screen.

12.1 Bank Dashboard

```
+-------------------------------------------------------------+
| RIPPLE | Dashboard | SMEs | Assessments | Reports | Profile  |
+-------------------------------------------------------------+
| Search SME: [_____________________________] [Search]         |
+---------------+---------------+---------------+-----------+
| Total SMEs    | Assessments   | High Risk     | Pending   |
|     248       |      391      |      42       |    17     |
+---------------+---------------+---------------+-----------+
| Risk Distribution             | Recent Assessments          |
| [chart]                       | ABC Ltd       High   18.7% |
|                               | XYZ Pvt Ltd   Low     4.2% |
+-------------------------------+-----------------------------+
| Data quality alerts / actions                               |
+-------------------------------------------------------------+
```

12.2 SME Search / Profile

```
+-------------------------------------------------------------+
| SME PROFILE: ABC Manufacturing Pvt Ltd                       |
+-------------------------------------------------------------+
| Basic Info       | Industry | Age | Location | Status        |
+-------------------------------------------------------------+
| Financial Summary                                            |
| Revenue | Profit | Debt | Cash | Receivables | Payables     |
+-------------------------------------------------------------+
| Alternative Data                                            |
| Payment | Transactions | Concentration | Trend | Reliability |
+-------------------------------------------------------------+
| Business Network                                             |
| [interactive graph preview]          [Open Full Network]    |
+-------------------------------------------------------------+
| [New Assessment] [View History] [Generate Report]           |
+-------------------------------------------------------------+
```

12.3 Data Upload / Input

```
+-------------------------------------------------------------+
| NEW ASSESSMENT — DATA INPUT                                 |
+-------------------------------------------------------------+
| Upload files                                                 |
| [Tax/GST File] [Financial File] [CSV] [Other]               |
|                                                             |
| Extraction status                                           |
| ✓ Revenue                ■XX,XX,XXX                         |
| ✓ Total Debt             ■X,XX,XXX                          |
| ! Payment history        Missing                            |
| ! Customer concentration Missing                            |
|                                                             |
| Manual fields required                                      |
| Payment history: [________________]                         |
| Customer concentration: [________________]                  |
|                                                             |
| Data completeness: 78%     Reliability: GOOD                |
| [Validate]                          [Continue Assessment]   |
+-------------------------------------------------------------+
```

12.4 Risk Result

```
+-------------------------------------------------------------+
| CREDIT RISK ASSESSMENT                                      |
+-------------------------------------------------------------+
| PD: 18.7%            RISK: MEDIUM-HIGH                      |
| Data confidence: GOOD                                       |
+-------------------------------------------------------------+
| Financial Analysis                                          |
| Liquidity | Leverage | Profitability | Coverage | Cash Flow |
```

```
+-------------------------------------------------------------+
| Why? — Top contributors                                     |
| ↑ Cash-flow volatility                                      |
| ↑ Customer concentration                                    |
| ↑ Payment delays                                            |
| ↓ Revenue stability                                         |
+-------------------------------------------------------------+
| Network Risk                                                |
| [graph]     Connected entities: 12                          |
|             High-risk exposures: 2                          |
+-------------------------------------------------------------+
| Scenarios: 9M | 12M | 18M | 24M                           |
+-------------------------------------------------------------+
| RIPPLE Recommendation: [Further Review]                     |
| Final Officer Decision: [Approve] [Reject] [Review]         |
+-------------------------------------------------------------+
```

12.5 SME Self-Service Result

```
+-------------------------------------------------------------+
| YOUR RIPPLE CREDIT READINESS REPORT                         |
+-------------------------------------------------------------+
| Indicative Score: 72/100       Risk Band: MEDIUM            |
+-------------------------------------------------------------+
| What is going well?                                         |
| ✓ Revenue growth                                            |
| ✓ Payment consistency                                       |
|                                                             |
| What needs attention?                                       |
| ! Cash-flow volatility                                      |
| ! High customer concentration                               |
| ! Missing financial information                             |
|                                                             |
| Improve your credit readiness                               |
| 1. Improve payment consistency                              |
| 2. Reduce concentration where feasible                      |
| 3. Maintain stronger financial records                      |
|                                                             |
| [Download Report]                                           |
+-------------------------------------------------------------+
```

13. Data Sources and Provenance

Potential data sources include public corporate/regulatory information, permitted tax/GST-related records, publicly available company/financial information, structured credit datasets, and controlled synthetic relationship data where real relationship data is unavailable.

For the prototype, the developer may use public datasets and synthetic/demo data. If synthetic relationships are used, the report must explicitly state that they are synthetic and do not represent real commercial relationships.

The system should maintain a simple provenance model: **source** → **extraction** → **validation** → **feature** → **model output**. The purpose is not to build a full enterprise data-governance platform, but to make it possible to explain where important inputs came from.

The project should prioritise reliable and verifiable data. A data source should not be included merely because it is publicly accessible. Relevance, reliability, legal/ethical suitability and potential bias must be considered.

14. CFA-Aligned Analytical Depth — What to Include and What Not to Overbuild

The request to use the CFA curriculum is best interpreted as a guide to **financial-analysis depth**, not as a requirement to implement every CFA topic. The current CFA curriculum spans areas including financial statement analysis, corporate issuers, fixed income/credit risk, quantitative methods, risk management, portfolio construction and ethics. ■cite■turn0search4■turn0search5■turn0search1■

CFA-related area	RIPPLE treatment
Financial Statement Analysis	Core: income statement, balance sheet, cash flow, ratios, trends, quality and consistency
Quantitative Methods	Core: statistics, probability, distributions, validation, calibration, scenario/sensitivity analysis
Corporate Issuers	Useful: business model, financing, governance and business/operating risk where data permits
Fixed Income / Credit Analysis	Core: creditworthiness, PD, leverage, liquidity, profitability, coverage, repayment capacity, credit risk
Risk Management	Core: scenario analysis, sensitivity, concentration/network exposure and uncertainty
Ethics	Core principle: transparency, human oversight, responsible use of financial data and avoidance of misleading auto
Equity valuation / derivatives / portfolio optimisation	Not central to the SME-loan prototype; include only if directly useful to a defined requirement
Level III portfolio construction	Not a direct system requirement; some risk-budgeting/stress concepts may inform future portfolio-level extensions

CFA Institute's credit-analysis material specifically discusses business-model assessment, financial statement analysis, cash-flow projections, profitability, leverage, coverage and probability of default; these are therefore appropriate anchors for RIPPLE's analytical layer. ■cite■turn0search10■

15. Bias, Redundancy and Model-Risk Controls

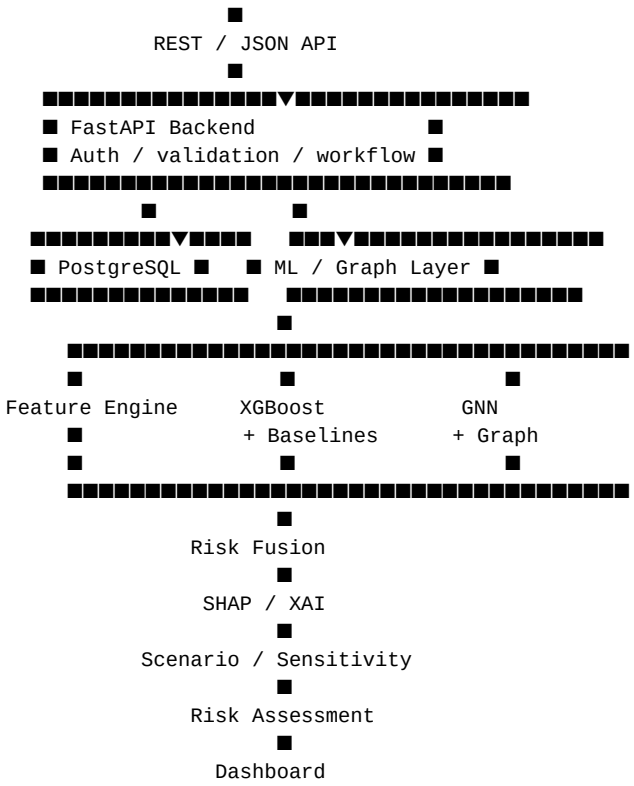
- **Do not maximise features blindly.** More data can increase noise, leakage and spurious correlations.
- **Control target leakage.** No feature should contain information that would only become available after the prediction point.
- **Check redundancy.** Highly correlated variables should be reviewed before modelling.
- **Separate train/test data properly.** Related businesses or repeated records must not leak across splits in a way that inflates performance.
- **Document synthetic data.** Synthetic graph relationships must not be presented as real-world observations.
- **Calibrate probabilities.** A PD should be useful as a probability, not just a ranking score.
- **Expose data quality.** Missing or unreliable inputs should be visible in the result.
- **Human oversight.** The model supports, not replaces, responsible lending judgement.
- **Fairness review.** Where sensitive/protected attributes are available, assess whether the model creates problematic disparities; do not use sensitive attributes merely because they improve prediction.

16. Non-Functional Requirements

Requirement	Expected behaviour
Usability	A bank analyst should understand the main risk result and drivers quickly without reading ML terminology.
Performance	Normal dashboard navigation and a saved assessment should feel responsive; exact latency target is Developer discretion.
Reliability	Failed extraction/model calls should produce clear errors rather than silent incorrect results.
Security	Protect uploaded financial information and restrict access by role as appropriate for the prototype.
Auditability	Store assessment timestamp, model/version and key input/output references.
Explainability	Every final risk result should have a human-readable explanation.
Maintainability	Separate frontend, backend, data pipeline, model and graph components.
Reproducibility	Model/data-processing versions should be recorded; Git/GitHub should be used.
Scalability	Architecture should not unnecessarily prevent future multi-SME portfolio analysis.

17. Suggested System Architecture





Recommended stack: React.js; FastAPI/Python; PostgreSQL; Pandas/NumPy/scikit-learn; XGBoost; NetworkX; PyTorch/PyTorch Geometric; SHAP; Git/GitHub; Docker. Exact libraries may change if the developer has a better justified choice.

18. Suggested Database Entities

Entity	Purpose
SME	Business identity and profile
FinancialRecord	Financial statement / financial-period data
AlternativeDataRecord	Behavioural/alternative signals and provenance
Relationship	Supplier/customer/business relationship
Assessment	One risk-assessment event for an SME
AssessmentInputSnapshot	Input state used for the assessment
Prediction	PD, risk band, model outputs and model version
Explanation	Feature contributions / human-readable explanation
ScenarioResult	Scenario/sensitivity outputs
DataQualityFlag	Missing, invalid, conflicting or low-confidence data
User	Bank/SME user identity and role if authentication is implemented

19. Developer-Discretion Areas

The following decisions are intentionally delegated to the developer. This is not an invitation to remove the required business functionality; it is freedom to choose the best technical implementation.

- Exact dataset(s), provided they are relevant, legally/ethically usable, sufficiently reliable, and documented.
- Exact target construction and modelling strategy, subject to meaningful validation.
- Prediction-horizon implementation, depending on the dataset's time structure.
- Graph node/edge schema and edge features.

- GNN architecture and training objective.
- Risk-fusion strategy.
- Missing-data imputation strategy.
- Feature-selection and dimensionality-reduction strategy.
- File parsing/extraction technology.
- Authentication implementation.
- Exact UI library/chart/graph library.
- Exact API route naming and internal software architecture.
- Whether GNN-specific explainability is implemented.
- Whether RIPPLE produces a direct approve/reject recommendation or a more conservative risk/review recommendation.

Every major discretionary choice should be recorded in a short **Architecture Decision Record** or project documentation note so the team can explain why it was chosen.

20. Phase II + Phase III Deliverables for 30 September

Area	Minimum expected output by milestone
SRS	Finalised requirements, users, features, data, scope and non-functional requirements
UML	Use-case, activity, class/domain, sequence and component/deployment diagrams as appropriate
Algorithm design	Data pipeline, feature engineering, XGBoost/baselines, GNN, fusion, SHAP, evaluation and scenario logic
Coding	Working frontend/backend/data/model modules
Database	PostgreSQL schema and working persistence
ML	Training/evaluation pipeline and saved model artefacts
Graph	Graph construction and working GNN prototype
XAI	Working SHAP explanation for tabular model
UI	Bank dashboard + assessment flow + result screen + SME self-service subset
Repository	GitHub repository with structured code, README, environment/setup instructions and version history

Definition of done: a developer should be able to clone the repository, follow the README, run the application, create/load an SME assessment, obtain a risk result, inspect the explanation and view the relevant business-network information using the demo/public/synthetic data supplied with the project.

21. Out of Scope for the Current Prototype

- Autonomous legally binding loan approval/rejection.
- Real-time production banking core-system integration.
- Direct access to private bank accounts without an approved integration.
- Production GST/UPI/banking integrations unless separately authorised and technically available.
- Use of confidential third-party financial information without permission.
- Claiming synthetic network relationships are real.
- Building a full enterprise credit-origination platform.
- Implementing every CFA curriculum topic irrespective of relevance.
- Adding exotic alternative data solely to make the project appear more advanced.

22. Future Scope

- **SME Financial Awareness & Credit Readiness:** personalised guidance on improving the drivers of creditworthiness.
- **Credit Portfolio Intelligence:** portfolio-wide concentration, clusters and interconnected exposure for lenders.
- **What-if Network Stress Testing:** simulate supplier/customer distress and evaluate network-level impact.
- **Real-time alternative-data integration:** authorised data-sharing ecosystems and continuous monitoring.
- **Counterfactual risk improvement:** estimate how changes in selected drivers could alter risk, with careful non-causal wording.
- **Federated credit intelligence:** collaborative model development without sharing raw SME data, subject to substantial future research.

23. Developer Checklist

1. Freeze the internal data schema and source/provenance convention.
2. Select and document the primary training/evaluation dataset(s).
3. Define the prediction target and prediction point without leakage.
4. Implement data ingestion and validation.
5. Implement financial/alternative feature engineering.
6. Implement baseline + XGBoost modelling.
7. Implement graph construction and GNN experiment.
8. Compare tabular and graph/combined approaches.
9. Implement SHAP for the tabular model.
10. Implement scenario/sensitivity views where supported by data.
11. Implement PostgreSQL persistence.
12. Implement FastAPI services.
13. Implement React bank workflow.
14. Implement SME self-service subset.
15. Implement report generation.
16. Add model/data version information.
17. Test missing/invalid/conflicting data paths.
18. Document limitations and synthetic-data assumptions.
19. Commit all code to GitHub with setup instructions.
20. Demonstrate the complete user journey end-to-end.

24. Final Build Principle

Do not build RIPPLE as "an XGBoost model with a dashboard." The intended system is a complete, explainable credit-risk workflow: reliable data ingestion → financial analysis → alternative-data analysis → business-network modelling → model comparison/fusion → PD/risk assessment → explanation → scenario analysis → human decision support.

At the same time, do not build a giant data-mining system simply because more data is available. Every important feature should have a reason, a source, a validation strategy and a clear relationship to credit risk. This balance—**depth without uncontrolled complexity**—is the central implementation principle.

Sources used for CFA-alignment: CFA Institute 2026 curriculum and official financial-analysis/credit-analysis materials. The CFA curriculum is used here selectively to strengthen credit-analysis depth; it is not treated as a complete software specification.

■cite■turn0search4■turn0search10■turn0search3■